

EPSY 5244: Survey Design, Sampling, and Implementation

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Reading materials

- Dillman, D. A., Smyth, J. D., & Christian, L. M. (2014). *Internet, Phone, Mail, and Mixed-Mode Surveys: The Tailored Design Method*. Wiley. Chapters 4-7.



Thinking About the Respondent's Perspective

- The underlying principle behind the best practices and guidelines for successful surveys is that the surveyor must get into a “respondent state of mind”
- We must consider what the questions, questionnaire, contacts, and so on will look like from the respondent's perspective



Issues to Consider When Starting to Craft Survey Questions

1. What concepts do I need to measure?

- Start by writing clear research questions that identify what will be studied

2. What type of information is the question asking for?

- Some information is readily available for respondents (e.g., demographics) whereas other types require respondents to think and/or recall (e.g., attitudes or behaviors) and are influenced by the context in which the question is asked
- Questions need to be designed so that they are not too difficult to answer and don't require respondents to recall details that are too specific



How many times have you read a magazine in the last six months?



Issues to Consider When Starting to Craft Survey Questions

3. What survey mode(s) will be used to ask the questions?

- Each mode has features that can enable some designs and constrain others:
 - Presence versus absence of an interviewer
 - Interviewer can provide extra motivation and assistance
 - But also may introduce *interviewer variance* and various forms of *interviewer bias*: *social desirability*, *acquiescence*, or when interviewer characteristics affect how people respond



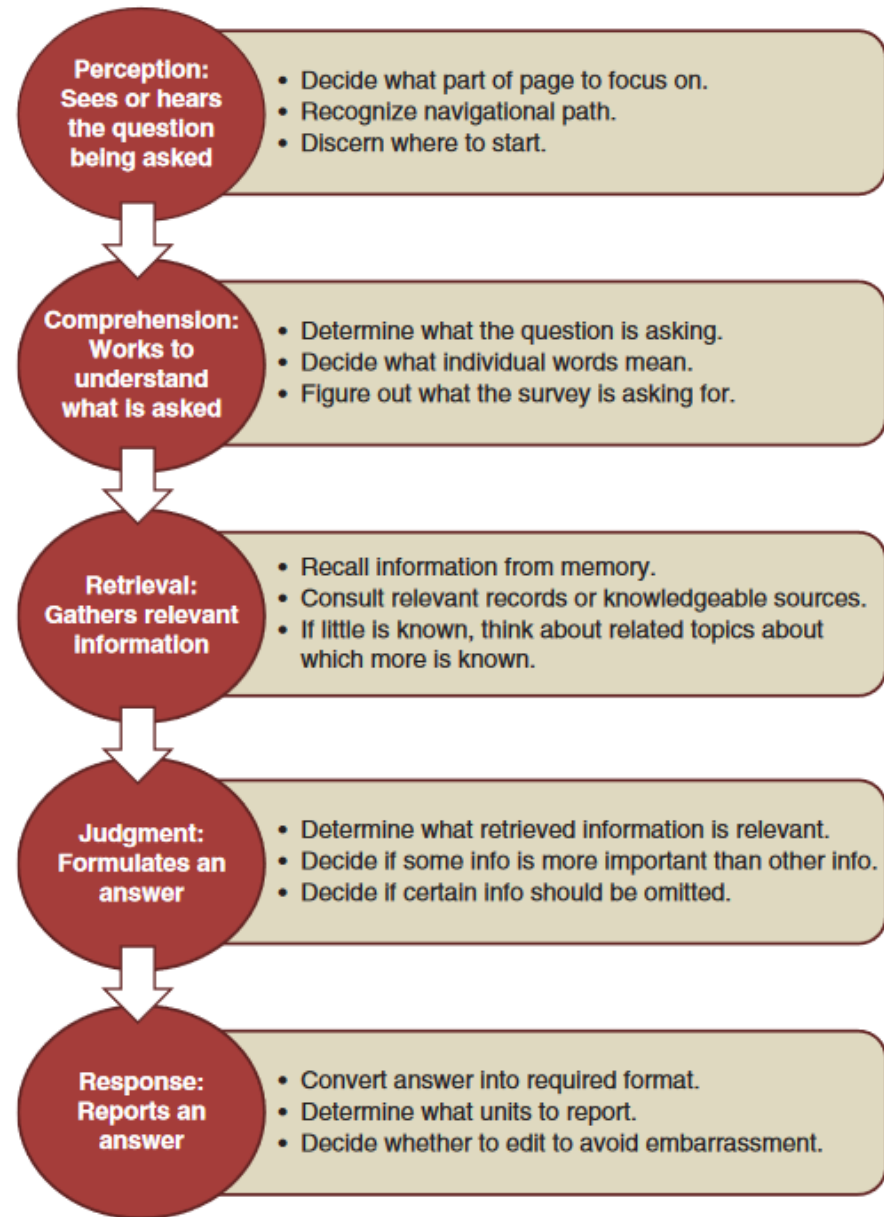
Issues to Consider When Starting to Craft Survey Questions

4. Is the question being repeated from another survey, and/or will answers be compared to previously collected data?
 - Consistency in how questions are asked is important for comparison
5. Will respondents be willing and motivated to answer accurately?
 - Think through the process respondents go through to help identify where things can go wrong



A common model of the survey response process

We want respondents to *optimize*: doing all the necessary mental work to provide quality answers, rather than *satisfice*: shortcut the response process



Approaches To Item Writing

Start from
scratch -
brainstorm

- Useful when no existing measures are available

Engage
w/potential
respondents or
experts

- Focus groups
- Interviews
- Literature

Use or adapt
existing
measures

- Most efficient (if possible)
- ***Can not*** re-use psychometric properties!



Anatomy of a Survey Question

- Vocabulary note: “*item*” vs. “*question*”
 - Item is a more general term – we are not always asking direct questions
- Question stem: the words that form the actual query itself
- Additional instructions: definitions or examples that will help respondents comprehend the meaning of the question or key concepts, or directions to further inform respondents how to respond
- Answer spaces or choices: where respondents report their response



Types of Question Formats

- *Open-ended*: allow respondents to formulate their answers how they want; offer a blank space or box where respondents write or type their own words (in self-administered surveys)
 - Should be used to collect rich, detailed information, or when little is known about the topic in advance
 - But require more work to analyze
 - And respondents are more likely to skip them

<i>Question stem</i>	What is the most important problem facing Nebraska today?
<i>Answer space</i>	
<i>Question stem with verbal and numeric instructions</i>	How many years have you lived in Nebraska? Please report only whole numbers. For example, if you have lived in Nebraska 20 months, please round to 2 years.
<i>Answer space with verbal and symbolic instruction</i>	Years lived in Nebraska

Types of Question Formats

- *Closed-ended:* respondents are provided with a list of answer choices from which they must choose a response
 - Important to ensure that all reasonable responses are offered

Closed-ended ordinal question

Question stem **Overall, how satisfied are you with living in Nebraska?**

Answer choices

- ☐ Completely satisfied
- ☐ Very satisfied
- ☐ Somewhat satisfied
- ☐ Not too satisfied
- ☐ Not at all satisfied

Closed-ended nominal question

Question stem **What is your current marital status?**

Answer choices

- ☐ Married
- ☐ Living with a partner
- ☐ Divorced
- ☐ Separated
- ☐ Widowed
- ☐ Never married



Types of Question Formats

- *Closed-ended*: respondents are provided with a list of answer choices from which they must choose a response
 - ▣ Important to ensure that all reasonable responses are offered

Partially closed-ended question

Question stem **What are your favorite women's sports at the University of Nebraska?**

Answer choices

- ☐ Basketball
- ☐ Cross Country
- ☐ Gymnastics
- ☐ Soccer
- ☐ Softball
- ☐ Swimming and Diving
- ☐ Tennis
- ☐ Volleyball
- ☐ Other: Please specify



Types of Question Formats

- Likert scale: When a Likert scale is used, the item is presented as a declarative sentence, followed by response options that indicate varying degrees of agreement with or endorsement of the statement.

	Strongly Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Strongly Agree
I feel good about my job	1	2	3	4	5
I get along well with others at work	1	2	3	4	5
I'm proud of my relationship with my supervisor at work	1	2	3	4	5
I can tell that other people at work are glad to have me there	1	2	3	4	5
I can tell that my co-workers respect me	1	2	3	4	5
I feel like I make a useful contribution at work	1	2	3	4	5



Types of Question Formats

- Semantic Differential scale is also called bipolar adjective rating scale. Respondents are asked to indicate their opinions or feelings toward a single statement using different pairs of adjectives framed as polar opposites (Bhattacharjee, 2012)

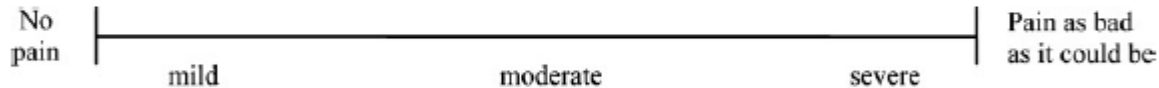
This session was:

bad	1	2	3	4	5	6	7	good
difficult	1	2	3	4	5	6	7	easy
valuable	1	2	3	4	5	6	7	worthless
shallow	1	2	3	4	5	6	7	deep
relaxed	1	2	3	4	5	6	7	tense
unpleasant	1	2	3	4	5	6	7	pleasant



Types of Question Formats

- Visual Analog scale presents the respondent with a continuous line between a pair of descriptors representing opposite ends of a continuum.
- If descriptive terms like 'mild', 'moderate', 'severe' or a numerical scale is added, it is called graphic rating scale.



Make sure the question applies to the respondent

- Every question should require an answer from every person of whom it is asked

Do you use your cell phone at least daily for voice calls, to access the Internet, or to send text messages?

- ☐ Yes
☐ No

Do you have a cell phone?

- ☐ Yes
☐ No

(If yes) Do you use your cell phone at least daily for each of the following?

	<u>Yes</u>	<u>No</u>
For voice calls	☐	☐
To access the Internet	☐	☐
To send text messages	☐	☐



Ask one question at a time

- Sometimes questions ask two (or more) things at a time
 - Called “Double-barreled” questions
- An easy mistake because we often want to limit the number of questions we ask in a survey

Has the district been responsive to your customer service needs with timely, informative, and accurate information?

Yes

Sometimes

No



Use Simple and Familiar Words

Replacing complex with simple words

Exhausted	Tired
Candid	Honest
Top priority	Most important
Leisure	Free time
Employment	Work
Courageous	Brave
Rectify	Correct

Replacing complex with simple phrases

Occupants of this household	People who live here
Your responses	Your answers
Postschool extracurricular activities	What you do after school
Subnational region	Area of the country



Use Simple and Familiar Words

Unnecessary use of specialized words and abbreviations

Have you filed Form SS-4 with the IRS?

- ☐ Yes
- ☐ No

A simplified revision

Have you filed an application for an employer identification number (Form SS-4) with the Internal Revenue Service?

- ☐ Yes
- ☐ No



Use Specific and Concrete Words to Specify the Concepts Clearly

Question with vague concepts

How many times did you eat together as a family last week?

☐ Number of meals

A revised question with more specific and concrete concepts

How many meals did you eat together as a family at home last week?

☐ Number of meals

A more specific revision

How many meals did you sit down to eat at home as a family last week?

☐ Number of meals



Use Complete Sentences That Take a Question Form, and Use Simple Sentence Structures

County vs. country

Common use of incomplete sentences

Number of years lived in Idaho

Years

Your city or town

City or Town

Your county

County

A revision using complete sentences

How many years have you lived in Idaho

Years

In what city or town do you live?

Name of City or Town

In what Idaho county do you live?

Name of Idaho County



Make Sure “Yes” Means Yes and “No” Means No

Questions should not include double negatives, or, in other words, require a respondent to say “yes” to mean no

A question containing a double negative

Do you favor or oppose not allowing the state to raise taxes without approval of 60% of the voters?

- ☐ Favor
- ☐ Oppose

A revision with no double negative

Do you favor or oppose requiring 60% approval by voters in order to raise state taxes?

- ☐ Favor
- ☐ Oppose

A revision that preserves important wording

In the September election you will be asked to vote on this referendum: “No state tax can be raised without approval by 60% of those voting in a statewide election.” If the election were held today, would you vote for or against approval?

- ☐ For
- ☐ Against



State both positive and negative sides in the question stem

Poor Designs

Do you favor congressional term limits of four years?

- ☐ Favor
- ☐ Oppose

Improved Designs

Do you favor or oppose congressional term limits of four years?

- ☐ Favor
- ☐ Oppose



Check all that apply

Check-all-that-apply formatted question

Which of the following items do you have? Please check all that apply.

- ☐ Desktop computer
- ☐ Laptop computer
- ☐ Cell phone
- ☐ E-reader
- ☐ Tablet computer
- ☐ iPod or MP3 player

A revision converting to the forced-choice format

Do you have each of the following items or not?

- | Yes | No | |
|-----------------------|-----------------------|--------------------|
| ☐ | ☐ | Desktop computer |
| ☐ | ☐ | Laptop computer |
| ☐ | ☐ | Cell phone |
| ☐ | ☐ | E-reader |
| ☐ | ☐ | Tablet computer |
| ☐ | ☐ | iPod or MP3 player |



Provide balanced scales with categories that are evenly spaced conceptually

Unbalanced scales with uneven distance between categories

How would you rate the quality of retail shops available to you in the Pullman/Moscow area?

- ☐ Excellent
- ☐ Very good
- ☐ Good
- ☐ Fair
- ☐ Poor

What do you think we should do about the country's defense budget?

- ☐ Increase it
- ☐ Keep it the same
- ☐ Decrease it a little
- ☐ Decrease it some
- ☐ Decrease it a lot

Balanced scales with more even distance between categories

How would you rate the quality of retail shops available to you in the Pullman/Moscow area?

- ☐ Very good
- ☐ Good
- ☐ Fair
- ☐ Poor
- ☐ Very poor

What do you think we should do about the country's defense budget?

- ☐ Increase it a lot
 - ☐ Increase it a little
 - ☐ Keep it the same
 - ☐ Decrease it a little
 - ☐ Decrease it a lot
-



Use clear language that leads to only one interpretation

Can you name more famous measurement scholars than Ben Wright?

Two possible interpretations:

Can you name measurement scholars *other* than Ben Wright?

Can you name measurement scholars *more famous than* Ben Wright?

(use one of these instead)



Don't make your respondents do math.

Calculations required:

*About how many hours
per month do you watch
television on
weekdays?*

Better:

*About how many hours
per weekday do you
spend watching
television?*



Visual Design Matters

Percent Responding
in Desired Format
(two-digit month, four-digit year)

Question 1 of 30
What month and year did you begin your studies at WSU?

Month Year

44%

Question 1 of 30
What month and year did you begin your studies at WSU?

Month Year

57%

Question 1 of 30
What month and year did you begin your studies at WSU? Please provide your answer using two digits for the month and four digits for the year.

Month Year

78%

Question 1 of 30
What month and year did you begin your studies at WSU? Please provide your answer using two digits for the month and four digits for the year.

MM YYYY

94%



Visual Design Matters

Which one of the following best **describes** the reason for your most *recent visit* to the Southgate Mall? ♥ **Shopping** for fun/entertainment ▷ Shopping for a needed item [] MALL WALKING/Exercise □ Other _____ ○ Dining at the mall ☆ Hanging out with friends Meeting new people ♣ *Conducting Business*

Which one of the following best describes the reason for your most recent visit to the Southgate Mall?

- ☐ Shopping for fun/entertainment
- ☐ Shopping for a needed item
- ☐ Mall walking/exercise
- ☐ Dining at the mall
- ☐ Hanging out with friends
- ☐ Meeting new people
- ☐ Conducting business
- ☐ Other



Visual Design Matters

Which one of the following best **describes** the reason for your most *recent visit* to the Southgate Mall? ♥ **Shopping** for fun/entertainment ▷ Shopping for a needed item [] MALL WALKING/Exercise □ Other _____ ○ Dining at the mall ☆ Hanging out with friends Meeting new people ♣ *Conducting Business*

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- ☐ Shopping for fun/entertainment
- ☐ Shopping for a needed item
- ☐ Mall walking/exercise
- ☐ Dining at the mall
- ☐ Hanging out with friends
- ☐ Meeting new people
- ☐ Conducting business
- ☐ Other



Visual Design Matters

An answer box that encourages respondents to give little information

Your answer to this question is very important for understanding what brings people to Washington State. Why did you choose to move to Washington State?

A revised answer box that encourages respondents to give more information

Your answer to this question is very important for understanding what brings people to Washington State. Why did you choose to move to Washington State?

Visual Design Matters

Uneven spacing of answer choices in an ordinal question

Compared to other veterinary services you have received, how would you rate the pricing at Pet Vet?

Very overpriced	Somewhat overpriced	Slightly overpriced	Average	Slightly underpriced	Somewhat underpriced	Very underpriced
☐	☐	☐	☐	☐	☐	☐

Revised even spacing of answer choices

Compared to other veterinary services you have received, how would you rate the pricing at Pet Vet?

Very overpriced	Somewhat overpriced	Slightly overpriced	Average	Slightly underpriced	Somewhat underpriced	Very underpriced
☐	☐	☐	☐	☐	☐	☐



Visual Design Matters

Conceptual and visual midpoints are properly aligned

To what extent do you approve or disapprove of the way Democrats in Congress are handling their job?

☐ Very much approve

☐ Somewhat approve

**Visual
midpoint**

☒ Neither approve nor disapprove

Conceptual midpoint

☐ Somewhat disapprove

☐ Very much disapprove



Visual Design Matters

Adding “don’t know” and “no opinion” categories misaligns the conceptual and visual midpoints

To what extent do you approve or disapprove of the way Democrats in Congress are handling their job?

☐ Very much approve

☐ Somewhat approve

☐ Neither approve nor disapprove **Conceptual midpoint**

Visual midpoint ☒ Somewhat disapprove

☐ Very much disapprove

☐ Don’t know

☐ No opinion



Visual Design Matters

Visually separating the nonsubstantive categories realigns the conceptual and visual midpoints

To what extent do you approve or disapprove of the way Democrats in Congress are handling their job?

☐ Very much approve

☐ Somewhat approve

Visual midpoint → ☒ Neither approve nor disapprove **Conceptual midpoint**

☐ Somewhat disapprove

☐ Very much disapprove

☐ Don't know

☐ No opinion



Exercise

Check if any of those items can be improved and, if yes, how.

1. How many days in January did you miss work because of illness or injury?		2. What are your job responsibilities?	
☐ Always		☐ Delivery	
☐ Most of the time		☐ Pick-up	
☐ Sometimes		☐ Invoicing	
☐ Rarely		☐ Loading and unloading	
☐ Never		☐ Customer service	
		☐ Maintenance	
3. Please check all of the descriptions you think describe your immediate supervisor.		4. Are you satisfied with your current job?	
	Yes	No	
Open-minded	☐	☐	☐ Very satisfied
Fair	☐	☐	☐ Somewhat satisfied
Honest	☐	☐	☐ Neither satisfied nor dissatisfied
Personable	☐	☐	☐ Somewhat dissatisfied
Productive	☐	☐	☐ Very dissatisfied
Expert	☐	☐	
5. Which of your university employees took research ethics training in 2013?			
a. Executive and administrative employees (e.g., deans, provosts, chairs, directors, etc.)	☐ All	☐ Most	☐ Some ☐ None
b. Tenure track faculty	☐ All	☐ Most	☐ Some ☐ None
c. Nontenure track faculty	☐ All	☐ Most	☐ Some ☐ None
d. Professional staff	☐ All	☐ Most	☐ Some ☐ None
e. Secretarial/clerical staff	☐ All	☐ Most	☐ Some ☐ None
f. Student employees	☐ All	☐ Most	☐ Some ☐ None

